从 Snapshot 副本还原文件

ONTAP 9

NetApp
January 31, 2022

This PDF was generated from https://docs.netapp.com/zh-cn/ontap/data-protection/snapshot-copies-work-concept.html on January 31, 2022. Always check docs.netapp.com for the latest.

目录

从 Snapshot 副本还原文件 1
从 NFS 或 CIFS/SMB 客户端上的 Snapshot 副本还原文件 1
从 Snapshot 副本还原单个文件 1
从 Snapshot 副本还原文件的部分内容 2
从 Snapshot 副本还原卷的内容 3

从 Snapshot 副本还原文件

从 NFS 或 CIFS/SMB 客户端上的 Snapshot 副本还原文件

NFS 或 CIFS/SMB 客户端上的用户可以直接从 Snapshot 副本还原文件，而无需存储系统管理员干预。

文件系统中的每个目录都包含一个名为 ` .snapshot` 的子目录， NFS 和 CIFS/SMB 用户可以访问该子目录。` .snapshot` 子目录包含与卷的 Snapshot 副本对应的子目录：

```
$ ls .snapshot
daily.2017-05-14_0013/              hourly.2017-05-15_1106/
daily.2017-05-15_0012/              hourly.2017-05-15_1206/
hourly.2017-05-15_1006/             hourly.2017-05-15_1306/
```

每个子目录都包含 Snapshot 副本引用的文件。如果用户意外删除或覆盖某个文件，他们可以将该文件从 Snapshot 子目录复制到读写目录，从而将该文件还原到父读写目录：

```
$ ls my.txt
ls: my.txt: No such file or directory
$ ls .snapshot
daily.2017-05-14_0013/              hourly.2017-05-15_1106/
daily.2017-05-15_0012/              hourly.2017-05-15_1206/
hourly.2017-05-15_1006/             hourly.2017-05-15_1306/
$ ls .snapshot/hourly.2017-05-15_1306/my.txt
my.txt
$ cp .snapshot/hourly.2017-05-15_1306/my.txt .
$ ls my.txt
my.txt
```

从 Snapshot 副本还原单个文件

您可以使用 `volume snapshot restore-file` 命令从 Snapshot 副本还原单个文件或 LUN 。如果您不想替换现有文件，则可以将此文件还原到父读写卷中的其他位置。

如果要还原现有 LUN ，则会创建一个 LUN 克隆，并以 Snapshot 副本的形式进行备份。在还原操作期间，您可以读取和写入 LUN 。

默认情况下，包含流的文件会进行还原。

步骤

1. 列出卷中的 Snapshot 副本：

   `volume snapshot show -vserver` *`svm`* `-volume` *`volum`*

有关完整的命令语法，请参见手册页。

以下示例显示了 `vol1` 中的 Snapshot 副本：

```
clus1::> volume snapshot show -vserver vs1 -volume vol1

Vserver Volume Snapshot                 State    Size  Total% Used%
------- ------ ---------- ----------- ------   -----  ------ -----
vs1  vol1   hourly.2013-01-25_0005  valid   224KB     0%    0%
               daily.2013-01-25_0010   valid   92KB      0%    0%
               hourly.2013-01-25_0105  valid   228KB     0%    0%
               hourly.2013-01-25_0205  valid   236KB     0%    0%
               hourly.2013-01-25_0305  valid   244KB     0%    0%
               hourly.2013-01-25_0405  valid   244KB     0%    0%
               hourly.2013-01-25_0505  valid   244KB     0%    0%

7 entries were displayed.
```

2. 从 Snapshot 副本还原文件：

   `volume snapshot restore-file -vserver` *`svm`* `-volume` *`volume`*`-snapshot` *`snapshot`* `-path` *`file_path`* `-restore-path` *`destination_path`*

   有关完整的命令语法，请参见手册页。

   以下示例将还原文件 `myfile.txt`：

```
cluster1::> volume snapshot restore-file -vserver vs0 -volume vol1
-snapshot daily.2013-01-25_0010 -path /myfile.txt
```

从 Snapshot 副本还原文件的部分内容

您可以使用 `volume snapshot partial-restore-file` 命令将一系列数据从 Snapshot 副本还原到 LUN 或 NFS 或 CIFS 容器文件，前提是您知道数据的起始字节偏移量和字节计数。您可以使用此命令还原在同一 LUN 中存储多个数据库的主机上的一个数据库。

步骤

1. 列出卷中的 Snapshot 副本：

   `volume snapshot show -vserver` *`svm`* `-volume` *`volum`*

   有关完整的命令语法，请参见手册页。

   以下示例显示了 `vol1` 中的 Snapshot 副本：

```
clus1::> volume snapshot show -vserver vs1 -volume vol1

Vserver Volume Snapshot                 State    Size  Total% Used%
------- ------ ---------- ----------- ------   -----  ------ -----
vs1  vol1   hourly.2013-01-25_0005  valid   224KB     0%    0%
               daily.2013-01-25_0010   valid   92KB      0%    0%
               hourly.2013-01-25_0105  valid   228KB     0%    0%
               hourly.2013-01-25_0205  valid   236KB     0%    0%
               hourly.2013-01-25_0305  valid   244KB     0%    0%
               hourly.2013-01-25_0405  valid   244KB     0%    0%
               hourly.2013-01-25_0505  valid   244KB     0%    0%

7 entries were displayed.
```

2. 从 Snapshot 副本还原文件的部分内容：

   `volume snapshot partial-restore-file -vserver` *`svm`* `-volume` *`volume`*`-snapshot` *`snapshot`* `-path` *`file_path`* `-start-byte` *`staring_byte`* `-byte-count` `_byte_count_count`

   起始字节偏移和字节计数必须是 4 ， 096 的倍数。

   以下示例将还原文件 `myfile.txt` 的前 4 ， 096 个字节：

```
cluster1::> volume snapshot partial-restore-file -vserver vs0 -volume
vol1 -snapshot daily.2013-01-25_0010 -path /myfile.txt -start-byte 0
-byte-count 4096
```

从 Snapshot 副本还原卷的内容

您可以使用 `volume snapshot restore` 命令从 Snapshot 副本还原卷的内容。

如果卷具有 SnapMirror 关系，请在从 Snapshot 副本还原后立即手动复制卷的所有镜像副本。否则，可能会导致镜像副本不可用，必须删除并重新创建这些副本。

步骤

1. 列出卷中的 Snapshot 副本：

   `volume snapshot show -vserver` *`svm`* `-volume` *`volum`*

   有关完整的命令语法，请参见手册页。

   以下示例显示了 `vol1` 中的 Snapshot 副本：

```
clus1::> volume snapshot show -vserver vs1 -volume vol1

Vserver Volume Snapshot                State    Size  Total% Used%
------- ------ ---------- ----------- ------   -----  ------ -----
vs1  vol1   hourly.2013-01-25_0005  valid   224KB     0%    0%
               daily.2013-01-25_0010   valid   92KB      0%    0%
               hourly.2013-01-25_0105  valid   228KB     0%    0%
               hourly.2013-01-25_0205  valid   236KB     0%    0%
               hourly.2013-01-25_0305  valid   244KB     0%    0%
               hourly.2013-01-25_0405  valid   244KB     0%    0%
               hourly.2013-01-25_0505  valid   244KB     0%    0%

7 entries were displayed.
```

2. 从 Snapshot 副本还原卷的内容：

   `volume snapshot restore -vserver` *`svm`* `-volume` *`volume" -snapshot _snapshot`*

   有关完整的命令语法，请参见手册页。

   以下示例将还原 `vol1` 的内容：

```
cluster1::> volume snapshot restore -vserver vs0 -volume vol1 -snapshot
daily.2013-01-25_0010
```

Copyright Information

Copyright © 2022 NetApp, Inc. All rights reserved. Printed in the U.S. No part of this document covered by copyright may be reproduced in any form or by any means-graphic, electronic, or mechanical, including photocopying, recording, taping, or storage in an electronic retrieval system- without prior written permission of the copyright owner.

Software derived from copyrighted NetApp material is subject to the following license and disclaimer:

THIS SOFTWARE IS PROVIDED BY NETAPP “AS IS” AND WITHOUT ANY EXPRESS OR IMPLIED WARRANTIES, INCLUDING, BUT NOT LIMITED TO, THE IMPLIED WARRANTIES OF MERCHANTABILITY AND FITNESS FOR A PARTICULAR PURPOSE, WHICH ARE HEREBY DISCLAIMED. IN NO EVENT SHALL NETAPP BE LIABLE FOR ANY DIRECT, INDIRECT, INCIDENTAL, SPECIAL, EXEMPLARY, OR CONSEQUENTIAL DAMAGES (INCLUDING, BUT NOT LIMITED TO, PROCUREMENT OF SUBSTITUTE GOODS OR SERVICES; LOSS OF USE, DATA, OR PROFITS; OR BUSINESS INTERRUPTION) HOWEVER CAUSED AND ON ANY THEORY OF LIABILITY, WHETHER IN CONTRACT, STRICT LIABILITY, OR TORT (INCLUDING NEGLIGENCE OR OTHERWISE) ARISING IN ANY WAY OUT OF THE USE OF THIS SOFTWARE, EVEN IF ADVISED OF THE POSSIBILITY OF SUCH DAMAGE.

NetApp reserves the right to change any products described herein at any time, and without notice. NetApp assumes no responsibility or liability arising from the use of products described herein, except as expressly agreed to in writing by NetApp. The use or purchase of this product does not convey a license under any patent rights, trademark rights, or any other intellectual property rights of NetApp.

The product described in this manual may be protected by one or more U.S. patents, foreign patents, or pending applications.

RESTRICTED RIGHTS LEGEND: Use, duplication, or disclosure by the government is subject to restrictions as set forth in subparagraph (c)(1)(ii) of the Rights in Technical Data and Computer Software clause at DFARS 252.277-7103 (October 1988) and FAR 52-227-19 (June 1987).

Trademark Information

NETAPP, the NETAPP logo, and the marks listed at http://www.netapp.com/TM are trademarks of NetApp, Inc. Other company and product names may be trademarks of their respective owners.